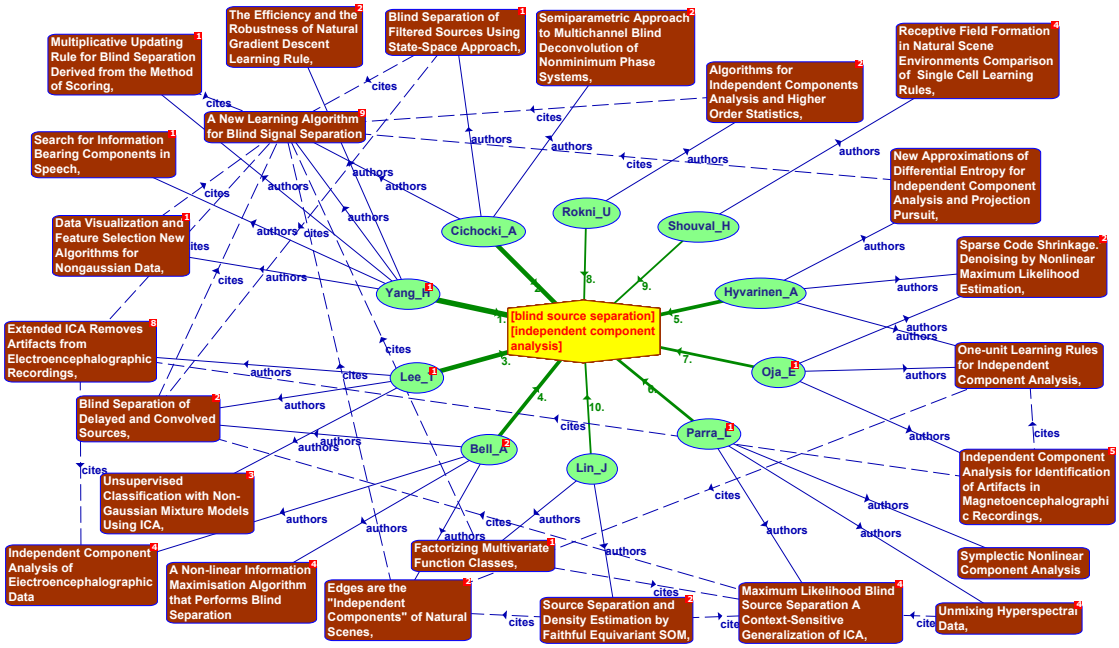


Figure 10.17: ETT example query in community browser.

Towards a community browser. Finally, models like ETT and the discussed extensions may be integrated with a graphical front-end that allows browsing of the community. Identifying expertise then becomes a navigation process in which queries are an inherent part. Motivated by the findings on representing community structure in Chapter 2, a proof-of-concept of such a front-end has been implemented as a graph-based interactive visualisation, extending earlier work in [Heinrich et al. 2005a, Heinrich 2005].

An exemplary screen exported from the browser is shown in Fig. 10.17, visualising the results of the tag query in Fig. 10.9(a). The distinct approach pursued here is to embed the query directly into the community graph and link it to the retrieved experts. Thus, the co-citation and authoring structure of the community relevant for the query (here the $authors(a, m)$ and $cites(m, m')$ relations of the AMQ network) becomes directly visible. In particular, the subgraph of maximum distance ("radius") 2 around the center node is displayed. Interactively, the searcher can view node details and annotate items of interest, collapse irrelevant items and move the center node to explore the community beyond the immediate query context.[18]

In addition to adding dedicated query nodes for term and tag queries as in the example, similarity queries can be issued from any node. This adds edges from the node to the most similar items according to the underlying model. As with term and tag queries, links to irrelevant query results (e.g., Shouval_H in the example) can be removed, which may be used to incorporate relevance feedback mechanisms on model level.

Using the ETT1 model in this browser, the retrieval functions (10.6) or (10.7) can be directly applied to link relevant authors to a query. To allow efficient retrieval of documents in addition,

[18] The numbers on the top-right of the nodes in Fig. 10.17 indicate hidden $authors(a, m)$ and $cites(m, m')$ links. In addition, the context radius may be varied, but graph views with radius >2 typically need additional filtering to curb visual complexity, e.g., showing only co-authors not in the results set (at radius 3).

parameters $\vec{\vartheta}_m$ for documents can be determined and cached. In ETT1, it is straight-forward to obtain these parameters by querying the model with document words and tags using (10.19).

Beyond the ETT models, especially incorporation of citation structure into the visual querying process is of interest. This may result in topic-specific links [Dietz et al. 2007] that may be also used to filter the links in the visualisation, additionally to show query-specific node importance based on models like TopicFlow [Nallapati et al. 2011], or to visually recommend relevant links between experts and articles using an approach based on RTM [Chang & Blei 2009].

In the browser, furthermore novel retrieval strategies may be integrated, such as [El-Arini & Guestrin 2011]. Here documents are identified by stating an initial set of relevant documents and then expanding this set by optimising an objective function based on document influence. In a user study, this method, which may be personalised with a searcher profile, outperformed state-of-the-art retrieval systems in a related-work search task.

10.6 Conclusions

In this chapter, the results of this thesis have been summoned to design topic models for virtual communities, in particular for an expert finding task in a scientific community that incorporates document tags. Central to this was the application of the NoMM development method from Chapter 9, starting with setting up the AMQ schema, making assumptions on the data and performing several iterations of model composition and predition. The designed models have been implemented using the Gibbs meta-sampler and tested against a sample corpus. This way, an end-to-end illustration of the design process has been given, validating the method by an exemplary case.

The expert–tag–topic models (ETT1–3) extend the state of the art towards retrieval of experts from documents with additional semantic tagging information or classification data. Experimentally, two of the models could be shown to outperform baseline models (LDA, ATM) in terms of retrieval performance and topic quality, which is attributed to the influence of additional tagging information. However, as the performance differences to the baselines are mostly not statistically significant, an extended study is indicated, based on larger data sets with more judged retrieval queries.

Based on the ETT models, an outline of model extensions has been given, including aspects of multiple thematic areas for experts, citation structure to capture author influence and thus improve retrieval results, as well as models of social tagging. Finally, a proof-of-concept visual community browser has been presented that combines ETT search with interactive exploration of the community's network structure.